CONNECTING PERSONALITY AND AFFECT

An implementation-oriented explanation of how two independent representations can be aligned through shared datasets, participant identity, emotion labels, and joint training.

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Executive summary

The target is a joint representation that integrates comparatively stable personality traits with context-dependent emotional state, while preserving their different meanings and timescales. A shared dataset is the bridge: it should connect personality measurements and emotion-bearing observations from the same identifiable participant or clearly identified character.

Personality traits P + Affective state E(t) + Context C(t)
↓
Identity-aware alignment and fusion → Joint state Z(t)
↓
Emotion transition / dialogue behavior prediction

What is established, and what is proposed?

Existing project description	Proposed integration
A questionnaire-based personality encoder-decoder; a provisional 16D geometry-focused personality representation.	A schema adapter for measured Big Five / IPIP outputs, with no assumption that five scores equal 300 item answers.
An affective model with an EmoBank CSV pilot and emotion-focused modeling.	Affect embeddings and optional VAD outputs aligned to identified dialogue turns.
A reusable scratch-trained GRU language backbone.	A joint fusion module and controlled baselines. The shared backbone is a possible downstream consumer.

Evidence boundary: the uploaded ManifoldLLm summary explicitly notes that model-level integration and behavioral conditioning had not been validated; this PDF describes a proposal, not a completed result.

1 | Why a connection is needed

Personality is a person-level property: questionnaire responses or trait scores aim to capture relatively stable individual differences. Emotion is an observation-level state: an utterance can express anger, joy, or another affective pattern depending on context and time. Training them separately does not demonstrate that their latent coordinates are compatible.

Property	Personality channel	Affect channel
Unit of analysis	Person / repeated survey occasion	Utterance / conversation turn / time point
Representative labels	IPIP items; Big Five traits	VAD ratings; emotion categories
Current illustrative data	IPIP-NEO-300 (licensed)	EmoBank (VAD)
Main integration risk	Missing item-level responses outside IPIP	Missing identities / personality labels in affect corpora

The minimum valid join

participant_id + observation_time + text / stimulus
↔ measured or annotated personality
↔ observed emotion / rated VAD (when available)

• Identity keys are essential: two unrelated rows from separate datasets cannot be joined just because their scores look similar.

• Repeated measures are valuable: a person with multiple utterances supports within-person changes and generalization tests.

• Measurement provenance matters: questionnaires, character annotations and model-generated affect scores are not equivalent ground truth.

2 | Dataset roles in the bridge

The expanded catalog contains 63 candidate resources or leads. This number is a discovery catalog, not 63 raw downloadable datasets or 63 verified participants-linked sources. Start with a few well-specified joint datasets, then use single-task datasets to strengthen the individual encoders.

Dataset	What it contributes	Role / caution
ETC Dataset	Participant-linked dialogue, personality questionnaire, and emotion descriptions	Strong candidate for person-level joint alignment; Japanese language.
CPED	Character-level personality, dialogue and emotion annotation	Excellent workflow benchmark; fictional characters, not questionnaire-grounded human identities.
PELD	Character personality and emotion transitions	Emotion transition benchmark; character annotations need provenance checks.
RealPersonaChat	Speaker identity, Big Five, multiple dialogue turns	Personality-dialogue grounding; emotion requires separate measurement or marked predictions.
EmoBank	Sentence-level continuous valence, arousal, dominance ratings	Affective encoder calibration; cannot directly join to personality identities.
GoEmotions / MELD	Emotion categories and contextual dialogue	Auxiliary training; emotion categories must not be treated as observed VAD without a validated mapping.

Prioritization rule

Tier 1: same-participant personality + text + observed affect. Tier 2: same-participant personality + text, with explicitly predicted affect. Tier 3: affect-only or personality-only corpora for separate pretraining. Tier 4: longitudinal and multimodal studies after fields, access, and modality adapters are verified.

Repository catalog: YinpingYin/datasets/catalog.csv and catalog.json. Follow original dataset access and licensing conditions; project metadata does not authorize republishing source corpora.

3 | Unified dataset contract

Normalize each authorized source into one auditable record layout. Missing variables must stay missing rather than being fabricated or automatically copied between unrelated people.

Field	Meaning and validation
dataset_id	Exact source / release identifier and original license reference.
participant_id	Source-local pseudonymous person identifier; never assumed globally shared.
utterance_id / timestamp	Unique observation and optional sequence ordering.
text / context	Utterance text and optionally adjacent dialogue context.
personality_type / values	Big Five scores, IPIP item responses, or annotations; record instrument and scale.
emotion_label / vad	Categorical emotion and/or observed VAD (when provided).
emotion_provenance	observed, annotated, predicted, or missing.
measurement_provenance	questionnaire, self-report, character annotation, or other source.
split_group	Participant/character grouping used to prevent train–test leakage.

Example record (illustrative, synthetic)

participant_id = demo_01; text = "I am excited about the results"; personality = {O: 0.68, C: 0.74, E: 0.62, A: 0.71, N: 0.31}; emotion_label = joy; emotion_provenance = annotated; timestamp = 2026-01-01T14:00. This is not a real participant record or evidence.

Adapter design

Each dataset receives a local parser and explicit mapping file. Do not train on mixed score scales before normalization. Attach source name and provenance to every label. Perform all quality checks and participant-level splits before constructing pairs for joint learning.

4 | Proposed model architecture

PERSONALITY INPUTS → Personality encoder f_P → z_P (stable / slow)

TEXT + CONTEXT → Affect encoder f_E → z_E(t) (dynamic / fast)

z_P, z_E(t), context → Fusion F → joint latent z(t)

Heads: emotion label / VAD prediction; next-state transition; dialogue behavior

The personality branch need not use the exact same numerical coordinates as the affect branch. The fusion network learns a relationship between them using paired, identity-consistent observations. The proposed decoder-induced manifold geometry can be evaluated after simple baselines; its superiority must not be assumed.

Practical input adapters

• IPIP item-level questionnaire → existing item encoder, if available. Big Five only → separate five-score adapter; do not infer 300 missing items.

• Emotion class annotations → categorical objective. True VAD labels → regression objective. Predicted VAD → optional auxiliary signal with explicit uncertainty.

• Language variation → multilingual text encoder or source-specific tokenizer; avoid treating Japanese and Chinese texts as English without a documented strategy.

Fusion choices in increasing complexity

Baseline A: affect-only; B: personality-only; C: concatenate standardized vectors + linear/MLP head; D: gated or attention-based fusion; E: geometry-aware fusion. Use matched splits, capacity control and confidence intervals for fair comparisons.

5 | Training, evaluation and failure modes

Staged training

Stage 1 — Separate: fit or reuse the existing personality and affect encoders on appropriate authorized data. Stage 2 — Align: use directly paired person-level observations for identity-aware learning. Stage 3 — Fuse: train a lightweight joint head before attempting geometric objectives. Stage 4 — Dynamic: add sequential emotion modeling only where temporal observations exist.

Illustrative objectives (not implemented claims)

$L = \lambda e \cdot L_emotion + \lambda v \cdot L_VAD + \lambda p \cdot L_personality + \lambda t \cdot L_transition + \lambda r \cdot L_regularization$. Each term is active only when its required observed/annotated supervision exists; pseudo-labels require separate reporting and weighting.

Evaluation checklist

Question	Evidence to collect
Does fusion improve prediction?	Emotion macro F1; VAD MAE / concordance (with ground truth only); paired baseline comparisons.
Does it generalize across people?	Hold out all utterances from test participants or independent characters; no leakage.
Is personality information used?	Personality shuffle test; remove-trait ablation; inspect per-trait sensitivity.
Is geometry beneficial?	Compare Euclidean and decoder-induced metrics under matched model capacity.
Is the state stable yet adaptable?	Between-person vs. within-person variation on repeated-observation datasets.

Important failure modes

Fabricated cross-dataset identity joins; character annotations presented as clinical psychometrics; categorical labels silently converted to VAD targets; mixing source-specific personality scales; evaluation leakage through speakers or dialogue duplicates; privacy or redistribution violations.

6 | Concrete implementation sequence

Milestone	Output	Acceptance criterion
M1. Dataset selection	One licensed joint dataset + schema mapping	Identity, personality, emotion provenance and usable splits documented.
M2. Data normalization	Validated standardized records	Parser passes missingness, scale and duplicate checks.
M3. Baselines	Emotion-only, personality-only, concatenated models	Held-out metrics and uncertainty reported.
M4. Fusion	Joint personality–affect module	Improvement demonstrated or honestly ruled out.
M5. Geometry + dynamics	Manifold and temporal ablations	At least one fair Euclidean comparison and one person-held-out test.

Questions to resolve in the actual codebase

• What precise tensor or CSV schema do the personality and affect encoders currently consume and emit?

• Which model checkpoints are available, and do they have stable feature dimensions?

• What licenses and participant constraints apply to the selected datasets?

• Is the first task emotion prediction, transition modeling, or language generation? Decide before selecting final losses.

Bottom line

The dataset bridge is a shared identity + label provenance + model-interface problem, not simply a file concatenation problem. A narrow, well-validated jointly labeled dataset can offer stronger evidence than dozens of disconnected corpora. Start with identity-aligned evaluation, then test whether manifold geometry adds measurable value.

Sources and status

Primary basis: the user-provided ManifoldLLm: Summary of Methods, Data and Results (2 pages), and the existing expanded 63-entry catalog / README. Representative source repositories: github.com/UEC-InabaLab/ETCDataset; github.com/scutcyr/CPED; github.com/preke/PELD; github.com/nu-dialogue/real-persona-chat; github.com/JULIELab/EmoBank. Dataset availability, licenses and exact columns should be checked at the upstream sources before use.